

Celebal Technologies Internship

Week 7 Assignment Report

Project: Delta Lake MERGE Implementation

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Course: Master of Computer Applications (MCA)

Tools: Python, Apache Spark (PySpark), Delta Lake, Databricks Community Edition, GitHub

1. Objective

The objective of this assignment was to build an incremental data processing pipeline using Apache Spark and Delta Lake. The project demonstrates how Delta Lake supports reliable updates and inserts using the MERGE operation while maintaining data consistency through ACID transactions.

2. Project Overview

A customer master dataset was loaded into Databricks from a CSV file. The dataset was inspected and cleaned by removing null values and duplicate records. The cleaned data was stored as a Delta Lake table. A second CSV file representing incremental customer data was then loaded. The Delta Lake MERGE command was used to update existing customer records and insert new records into the Delta table. Finally, validation steps were performed to verify record counts and ensure there were no duplicate customer records.

3. Technologies Used

Python for programming, Apache Spark (PySpark) for distributed processing, Delta Lake for transactional storage and MERGE operations, Databricks Community Edition for development, CSV as the source format, and GitHub for version control and project submission.

4. Implementation Workflow

- Step 1: Configure Spark with Delta Lake support.
- Step 2: Load the customer master dataset from CSV.
- Step 3: Display schema and preview the data.
- Step 4: Remove null values and duplicate records.
- Step 5: Save the cleaned data as a Delta table.
- Step 6: Load the incremental customer dataset.
- Step 7: Perform the MERGE operation to update matching records and insert new records.
- Step 8: Validate the final dataset by checking total records and duplicate Customer IDs.
- Step 9: Display the final output and stop the Spark session.

5. Key Features

Incremental data processing, Delta Lake MERGE, ACID-compliant storage, Spark DataFrame transformations, data cleaning, schema validation, duplicate detection, and scalable data management.

6. Results

The MERGE operation completed successfully. Existing customer records were updated, new customer records were inserted, and the final dataset was validated. The project demonstrates a

practical data engineering workflow suitable for handling continuously arriving business data.

7. Learning Outcomes

This assignment strengthened practical knowledge of Apache Spark, Delta Lake, DataFrame operations, incremental ETL pipelines, and version-controlled project organization using GitHub.

8. Conclusion

The project successfully implemented a complete Delta Lake MERGE pipeline. It demonstrated how Spark and Delta Lake can efficiently process incremental data while maintaining data quality and consistency. The solution follows a workflow commonly used in modern data engineering projects.